4.4.3 Superposition

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	**(i)** the principle of superposition of waves	
	(ii) techniques and procedures used for superposition experiments using sound, light and microwaves	**PAG5**
(b)	graphical methods to illustrate the principle of superposition	
(c)	interference, coherence, path difference and phase difference	
(d)	constructive interference and destructive interference in terms of path difference and phase difference	
(e)	two-source interference with sound and microwaves	
(f)	Young double-slit experiment using visible light	Learners should understand that this experiment gave a classical confirmation of the wave-nature of light. HSW7 Internet research on the ideas of Newton and Huygens about the nature of light.
(g)	**(i)** $\lambda = \frac{ax}{D}$ for all waves where $a \ll D$	*M4.6*
	(ii) techniques and procedures used to determine the wavelength of light using (1) a double-slit, and (2) a diffraction grating.	**PAG5** $d \sin\theta = n\lambda$ and diffraction gratings will only be assessed at A level